

Statistical Methods Quiz #3 (50 minutes)

'26. 05. 22. (Friday) Cadet number(교번): _____ Name: _____ Score: _____

<Instructions>

1. It is open-book, closed-web test.
 2. You are not allowed to discuss on the problems during the exam.
 3. Import *school.csv* data first. (Use all the string variables as factors.)
- sex:** Gender. F = female, M = male
age: Age. Numeric values ranging from 15 to 22
Pstatus: Parents' cohabitation status. T = living together, A = apart
Medu: Mother's education level. 0 = none, 1 = primary education, 2 = grades 5-9, 3 = secondary education, 4 = higher education
Fedu: Father's education level. 0 = none, 1 = primary education, 2 = grades 5-9, 3 = secondary education, 4 = higher education
guardian: Primary guardian. Mother, father, or other
traveltime: Travel time to and from school. 1 = less than 15 min, 2 = 15-30 min, 3 = 30 min-1 hour, 4 = more than 1 hour
studytime: Study time. 1 = less than 2 hours, 2 = 2-5 hours, 3 = 5-10 hours, 4 = more than 10 hours
failures: Number of past class failures. 1, 2, 3, or 4+
freetime: Free time. Ordinal value from 1 (very low) to 5 (very high)
famrel: Family relationship quality. Ordinal value from 1 (very bad) to 5 (excellent)

1. Import *school.csv* dataset. (Activate "Strings as factors" option) Use the **str** function to examine the structure of the *school* data. How many observations and how many variables does the dataset contain? [15 points]

<R-code>

```
str(school)
```

<Output>

```
'data.frame':      384 obs. of  12 variables:
 $ sex       : Factor w/ 2 levels "F","M": 1 1 1 1 2 2 1 2 2 1 ...
 $ age       : int  18 17 15 15 16 16 17 15 15 15 ...
 $ Pstatus   : Factor w/ 2 levels "A","T": 1 2 2 2 2 2 1 1 2 2 ...
 $ Fedu      : int   4 1 1 2 3 2 4 2 4 4 ...
 $ Medu      : int   4 1 1 4 4 2 4 3 3 4 ...
 $ guardian  : Factor w/ 3 levels "father","mother",...: 2 1 2 2 2 2 2 2 2 2 ...
 $ studytime : int   2 2 2 3 2 2 2 2 2 2 ...
 $ traveltime: int   2 1 1 1 1 1 2 1 1 1 ...
 $ failures  : int   0 0 3 0 0 0 0 0 0 0 ...
 $ famrel    : int   4 5 4 3 5 4 4 4 5 3 ...
 $ freetime  : int   3 3 3 2 4 4 1 2 5 3 ...
 $ absences  : int   6 4 10 2 10 0 6 0 0 0 ...
```

The number of observations: 384

The number of variables: 12

2. If a student's "absences" variable is larger than 9, the student is classified as being "lazy". Define a factor variable "lazy" in *school* dataset using **ifelse** and **factor** function. (with levels "Yes" and "No") [15 points]

<R-code>

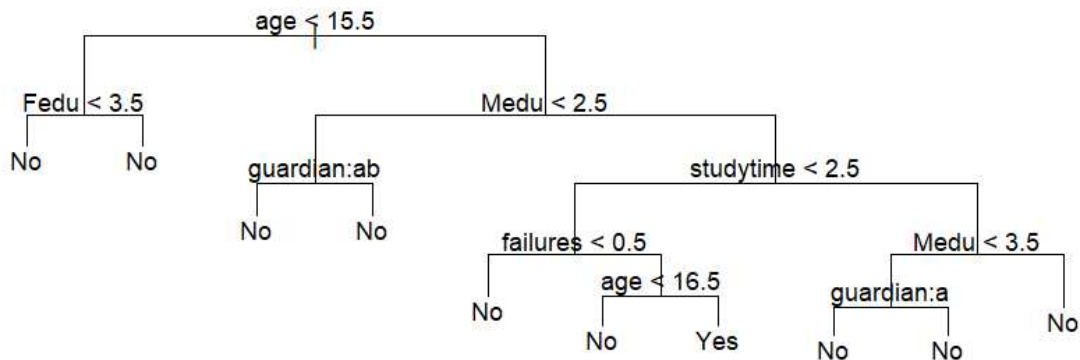
```
school$lazy = factor(ifelse(school$absences > 9, "Yes", "No"))
```

3. Using the *school* data, fit a **classification tree** with **lazy** as the response variable and all other variables **except absences** as explanatory variables. Also, visualize the fitted tree. What is the variable used in the first split? [15 points]

<R-code>

```
library(tree)
model = tree(lazy ~ . -absences, data = school)
plot(model)
text(model)
```

<Output (plot)>



The variable used in the first split: age

4. Using the *school* data, obtain the predicted response variable. Then, create a **confusion matrix** and compute the **misclassification error rate**. [20 points]

<R-code>

```
pred = predict(model, school, type = "class")
true = school$lazy
table(pred, true)
mean(pred != true); summary(model)
```

<Output>

```
> table(pred, true)
      true
pred  No  Yes
  No 297  68
  Yes   5  14
> mean(pred != true)
[1] 0.1901042
```

Misclassification error rate: 0.1901042

5. Using the school data, fit a **random forest** model with **lazy** as the response variable and all other variables **except absences** as explanatory variables. (Set the number of variables randomly selected at each split to **3**.) Using the data, obtain the predicted response variable. Then, create a **confusion matrix** and compute the **misclassification error rate**. Based on misclassification error rate, which model performs better for the training data: the **single decision tree** or the **random forest**? (Select one.) [20 points]

<R-code>

```
library(randomForest)
model2 = randomForest(lazy ~ . -absences, data = school,
                      mtry = 3)
pred = predict(model2, school, type = "class")
true = school$lazy
table(pred, true)
mean(pred != true)
```

<Output>

```
> table(pred, true)
      true
pred  No Yes
No   302  3
Yes    0 79
> mean(pred != true)
[1] 0.0078125
```

Misclassification error rate: 0.0078125

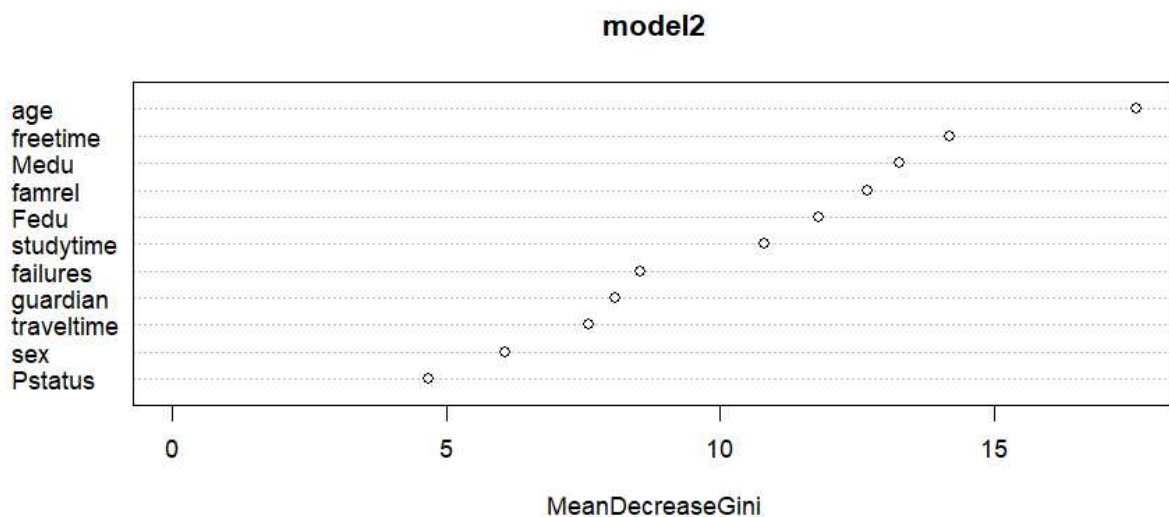
The ([single decision tree](#) / [random forest](#)) model has better performance.

6. For the random forest model, visualize the **variable importance**. Which variable is the most important? [15 points]

<R-code>

```
varImpPlot(model2)
```

<Output (plot)>



The most important variable: age